LAGOS BUSINESS SCHOOL · MEMBA 14 · EEGB · PART II

SESSION 6

AD–AS in a Shock-Prone World

From markets, to macro, to decisions that matter — in Nigeria and the world.



Reading: Olivier Blanchard, Macroeconomics · Facilitator: Bongo Adi

Microeconomics in Four Ideas

Prices ration	Demand, supply and elasticity decide how much of a cost shock you can pass on.
Forecast honestly	Trend, seasonality and scenarios beat a single 'confident' number.
Structure matters	Platforms, network effects and switching costs shape who earns the profit.
Markets can fail	Externalities and subsidies change behaviour — and someone always pays.

TODAY'S PLAN

Session 6 at a Glance — 75 Minutes

Why prices across the whole economy rise — and why the cause decides the cure.



Hook

Nigeria's 2023–24 inflation surge

The AD–AS Model

Demand shocks, supply shocks and what each does to output

Measuring Inflation

Headline, food, core and the trimmed mean

Expectations & Lab

Phillips curve, persistence, and the inflation diagnosis lab

BY THE END OF TODAY

Session 6 Learning Objectives

You will be able to:

01

↗

Use the AD–AS Model

Tell a demand shock from a supply shock, and predict what each does to prices and output.

02

±

Read Inflation Measures

Explain headline, food, core and trimmed-mean inflation, and why each answers a different question.

03

✦

Respect Expectations

Explain why inflation is sticky, and why central banks care so much about what people expect.

THE HOOK

From 22% to 34% in Eighteen Months

✦

“Prices are rising because people have too much money. Raise interest rates and it will stop.”

It sounds plausible. But in 2023–24 three different forces hit Nigerian prices at once:

Petrol subsidy removed

In May 2023 the pump price roughly tripled within weeks. Transport and food costs followed.

⇄ The naira devalued

After the June 2023 unification, the naira lost well over half its dollar value. Imported goods and inputs repriced.

$ Money and credit grew

Broad money and credit grew very fast, partly because dollar balances were revalued at the weaker rate.

NIGERIAN BUSINESS CASE

Demand Problem, Supply Problem — or Both?

The situation

After the May–June 2023 reforms, Nigerian inflation climbed from about 22% to the mid-30s by 2024. The CBN raised its policy rate sharply. Firms faced rising costs and weakening customers at the same time.

The decision question

Was the surge mainly a **demand** shock (too much spending) or a **supply** shock (costs pushed up)? The answer decides whether higher interest rates are the right medicine.

The data we have

ng_macro_monthly.csv — headline, food and core inflation, exchange rates, policy rate and PMI, 2015–2025. *Stylised teaching data.* We use 2015–2024 for like-for-like comparisons because the CPI was rebased in 2025.

What a good answer looks like

A short **diagnosis** with evidence (food vs core, exchange-rate pass-through, activity), and what it implies for policy and for **your pricing**.

PART

01

The AD–AS Model

One diagram for the whole economy.

Aggregate Demand and Aggregate Supply

Aggregate demand (AD)

Total spending in the economy

- Households, firms, government and the rest of the world
- Shifts right when credit is cheap, budgets expand or exports boom
- Shifts left when rates rise or confidence falls
- Policy levers: interest rates and the budget

Aggregate supply (AS)

What firms can produce at each price level

- Depends on costs: energy, imported inputs, wages, logistics
- Shifts up/left when costs jump (fuel, FX, insecurity on farms)
- Shifts down/right with productivity, power and better roads
- Policy levers: slow, structural reforms

*Where the two curves cross sets the **price level** and **real output**. Shocks move one curve or both.*

Demand Shock vs Supply Shock

Same rise in prices, opposite news for output

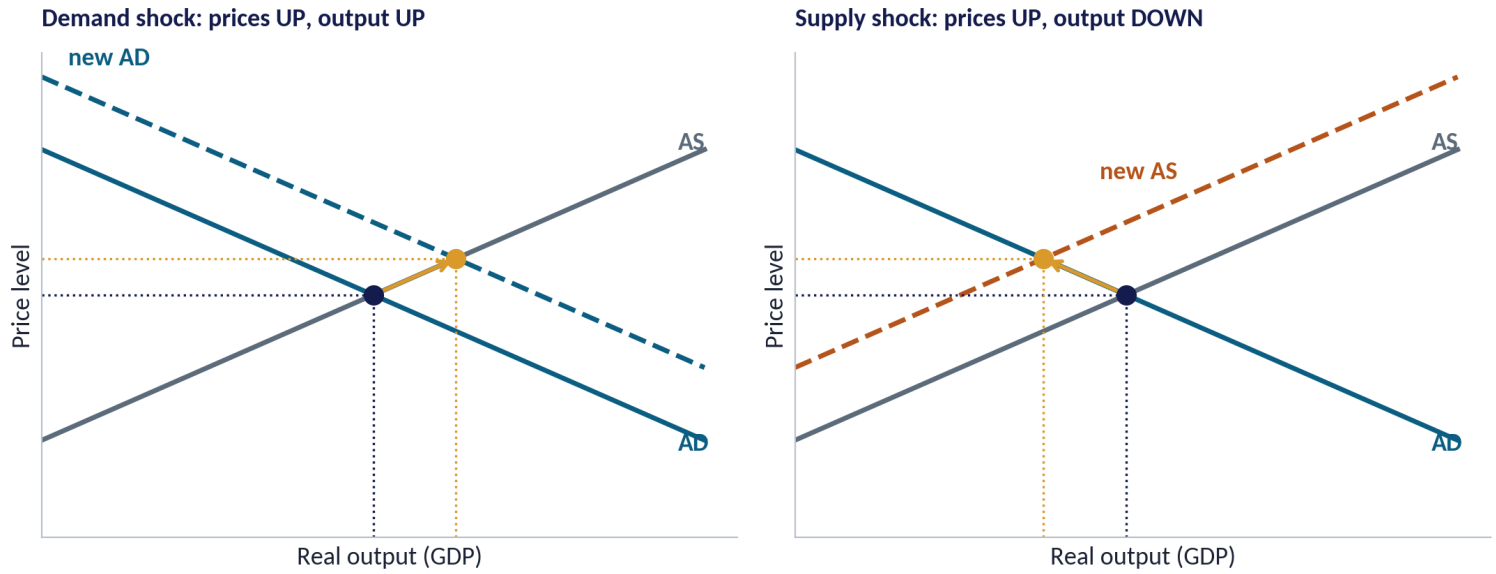


Fig S06-F01 · replicate: s06_figures.R / .py

WHAT TO SEE

- **Demand shock**: spending jumps → prices **and** output rise together. Overheating.
- **Supply shock**: costs jump → prices rise but output **falls**. Stagflation.
- Both raise inflation, so inflation alone cannot tell you which happened.
- Look at **output and activity** too: are firms selling more, or less?

The Same Medicine Does Not Cure Both

IN PLAIN ENGLISH

Higher interest rates work on **demand**: they make borrowing dearer and cool spending. Against a **supply** shock they cannot bring fuel or dollars back — they mainly reduce output further, while stopping the shock from spreading into wages and expectations.

EXAMPLE

Mid-2023 to end-2024, average inflation was **30.5%** against **19.6%** in the eighteen months before, while the PMI slipped from **52.7** to **49.4**. Prices up, activity down: the supply-shock signature.

THE FORMULA

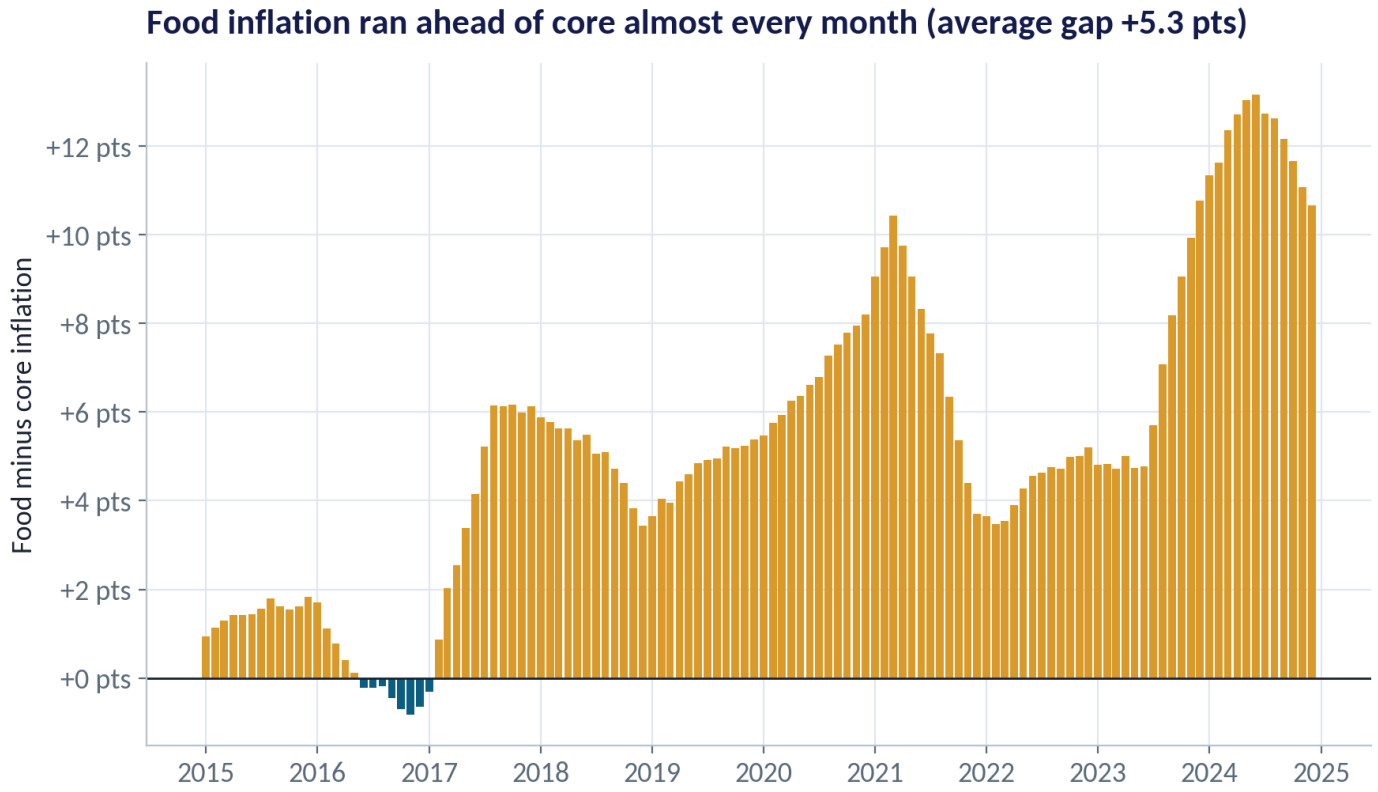
demand shock: P↑ Y↑ · supply shock: P↑ Y↓

WHY A MANAGER CARES

If you misdiagnose a supply shock as demand, you expect rate rises to fix prices quickly. They won't — so plan for costs that stay high and customers who spend less.

EVIDENCE 1

Food Running Ahead of Core



Source: EEGB stylised teaching data (data/). Not official statistics.

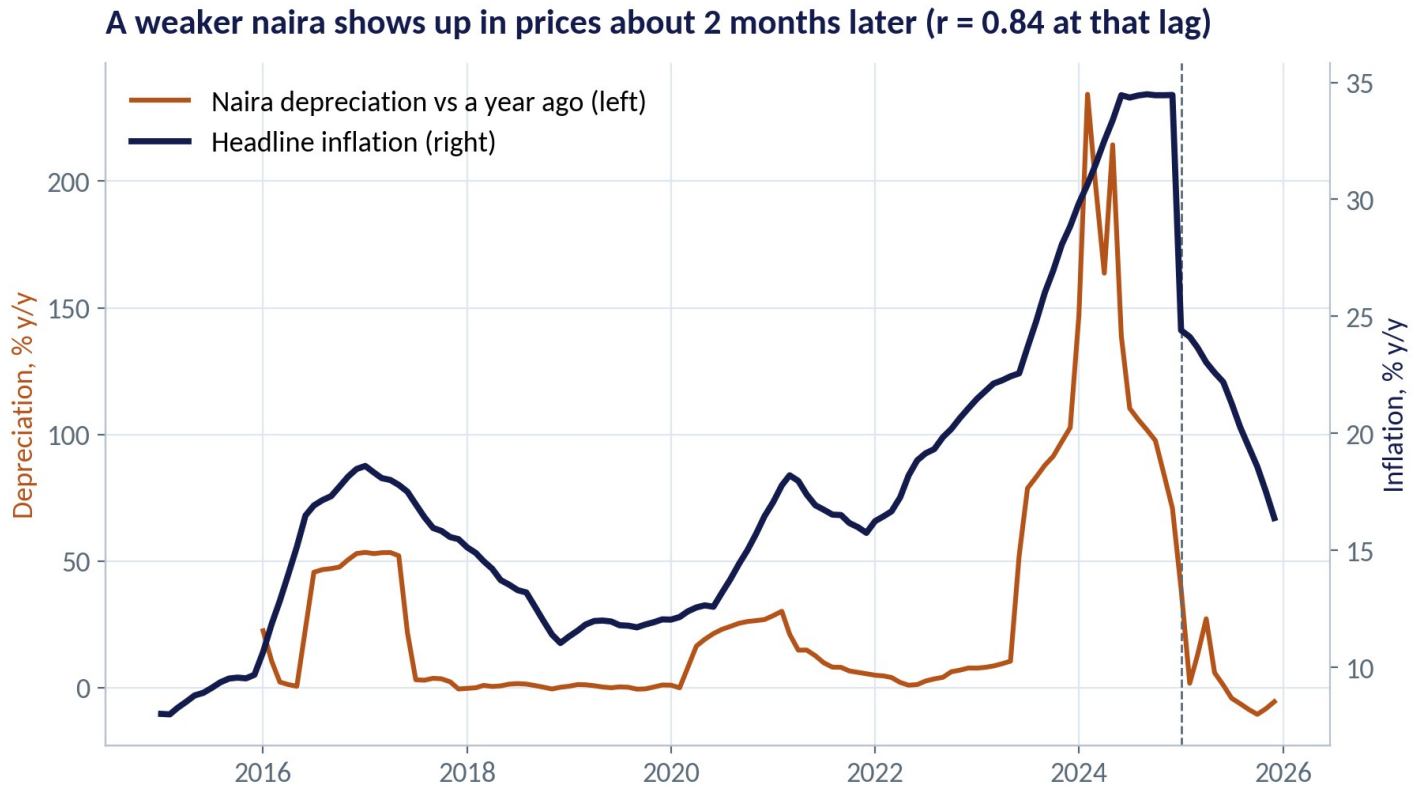
Fig S06-F02 · replicate: s06_figures.R / .py

WHAT TO SEE

- **Food** inflation exceeded **core** inflation almost every month.
- The gap averaged **+5.3 points** over 2015–2024 and widened to about **+12 points** in 2024.
- Food prices react to fuel, transport, insecurity and the exchange rate — mostly **supply** forces.
- A wide food–core gap is a warning that interest rates alone will not fix the problem.

EVIDENCE 2

Exchange-Rate Pass-Through



Source: EEGB stylised teaching data (data/). Not official statistics.

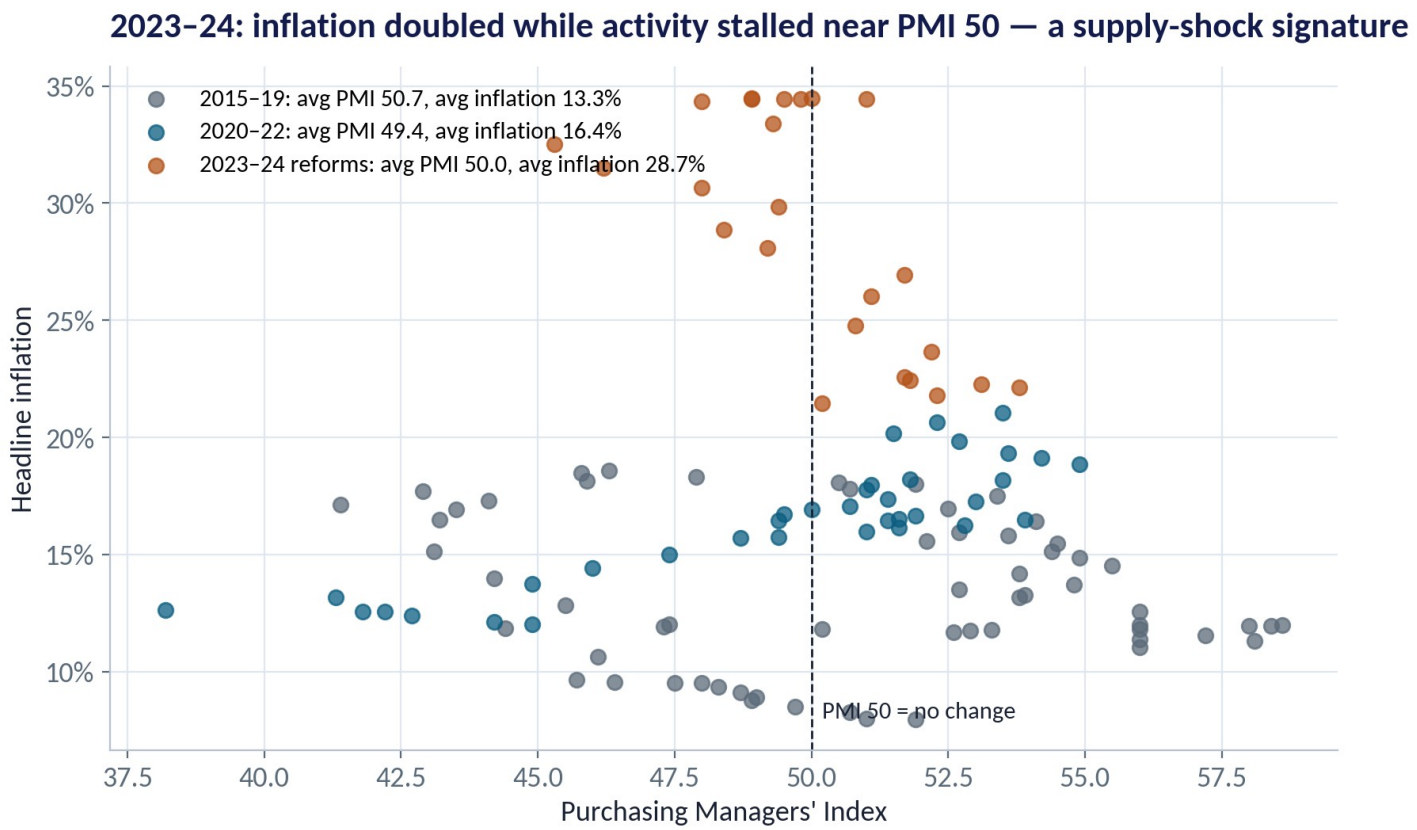
Fig S06-F03 · replicate: s06_figures.R / .py

WHAT TO SEE

- When the naira weakens, imported goods and inputs cost more in naira.
- In this data the link is strongest about **2 months** later (correlation **0.84**).
- The naira's fall after mid-2023 was the single largest cost shock for many firms.
- Pass-through is a **supply** shock to importers — but a weaker naira also helps exporters.

EVIDENCE 3

Prices Up, Activity Stalled



Source: EEGB stylised teaching data (data/). Not official statistics.

Fig S06-F04 · replicate: s06_figures.R / .py

WHAT TO SEE

- Each dot is one month: PMI (activity) against inflation.
- In 2015–19 inflation averaged **13.3%** with the PMI around **50.7**.
- In 2023–24 inflation averaged **28.7%** while the PMI averaged only **50.0**.
- Inflation more than doubled without a boom in activity — consistent with a supply shock.

Quick Check: What Would You Say?

?

A sudden rise in diesel and imported-input costs pushes Nigerian prices up while factories cut output. In AD–AS terms this is:

- **A** A rightward shift of aggregate demand
- **B** An upward (leftward) shift of aggregate supply
- **C** A movement along aggregate demand caused by higher incomes
- **D** A fall in aggregate demand caused by lower prices

Discuss with your neighbour for 60 seconds — answer and reasoning in the speaker notes.

PART

02

Measuring Inflation

Which number answers which question?

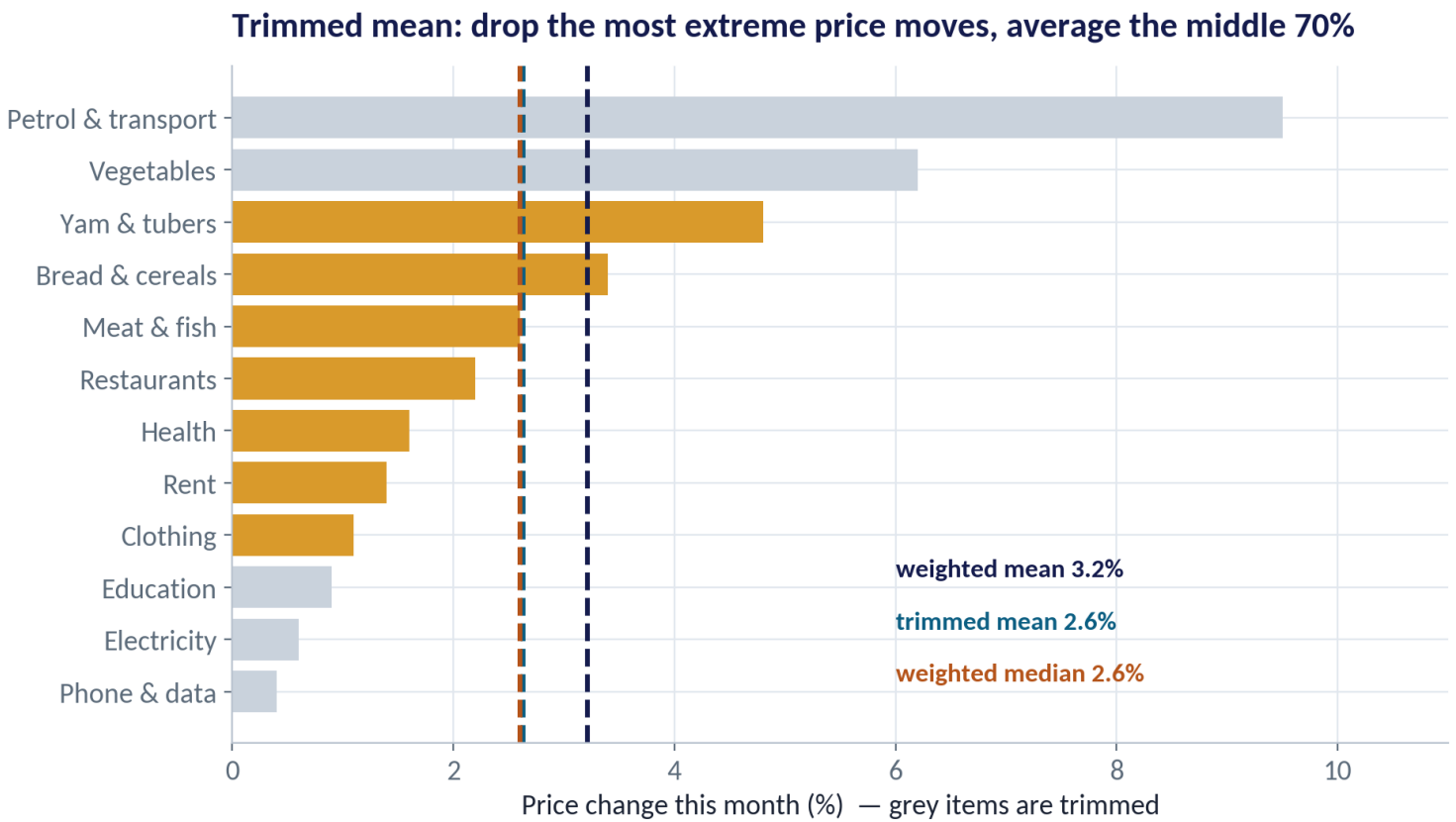
FOUR MEASURES

Headline, Food, Core and Trimmed Mean

Measure	What it includes	Question it answers	Weakness
Headline CPI	The full consumer basket	How fast is the cost of living rising?	Swings with food and fuel shocks
Food	Food items in the basket	How hard are households being squeezed?	Weather, insecurity and transport drive it
Core	Excludes farm produce and energy	What is the underlying trend?	Excludes what people actually pay for
Trimmed mean	Drops the most extreme price changes each month	What is the typical price change?	Less familiar; not always published

*Use **headline** for wages and contracts, **core or trimmed mean** to judge whether inflation is truly slowing.*

Taking Out the Extremes



Illustrative basket generated in the script; not NBS data.

Fig S06-F05 · replicate: s06_figures.R / .py

WHAT TO SEE

- This illustrative month: petrol and vegetables jumped; data and electricity barely moved.
- The **weighted mean** (headline) is **3.2%** — pulled up by two big movers.
- Drop the top and bottom 15% of the basket by weight: the **trimmed mean** is **2.6%**.
- When the two diverge, a few items are driving headline inflation — often a supply story.

A WARNING ABOUT YARDSTICKS

Rebasing Changes the Ruler

IN PLAIN ENGLISH

A price index compares today's basket with a **base year**. When statisticians **rebase**, they update the base year and the basket weights. Year-on-year rates before and after are **not** directly comparable.

EXAMPLE

Nigeria's CPI was rebased in January 2025. In this dataset headline inflation drops from about **34%** in December 2024 to about **24%** in January 2025 — much of that is the **new ruler**, not a sudden collapse in price pressure.

THE FORMULA

$$\text{inflation} = (\text{CPI}_t \div \text{CPI}_{t-12} - 1) \times 100$$

WHY A MANAGER CARES

Before you change prices, wage offers or contract indexation clauses, check which index and which base year the number uses.

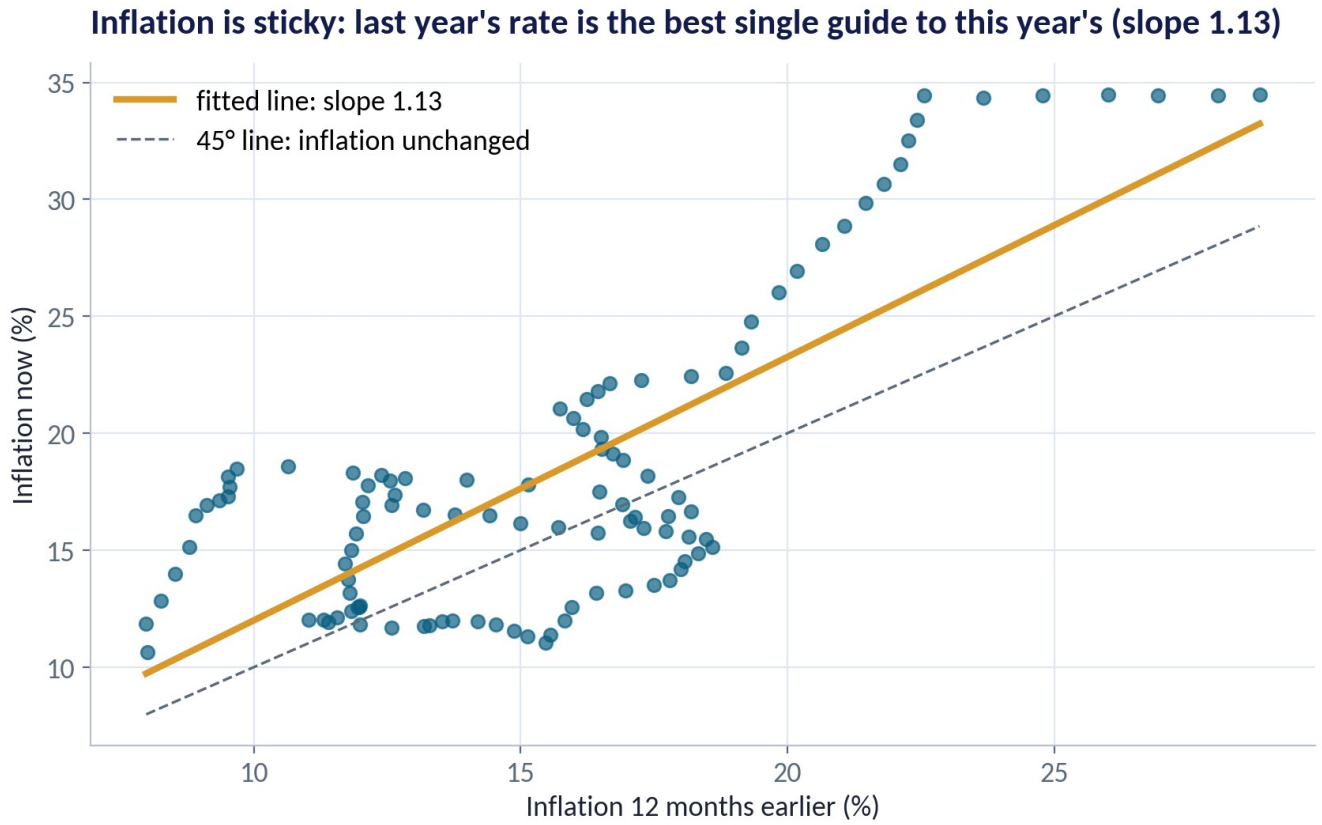
PART

03

Expectations and Persistence

Why inflation is easy to start and hard to stop.

Inflation Is Sticky



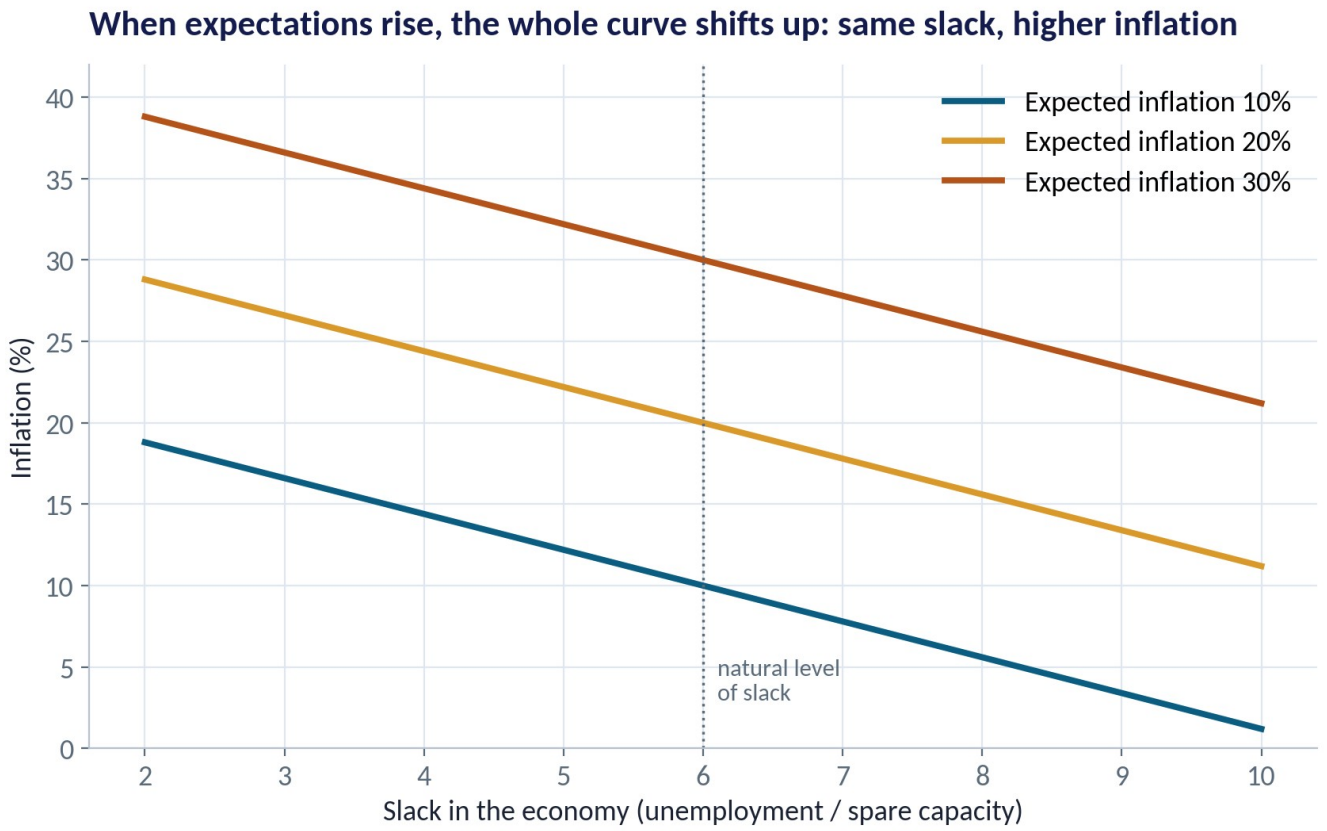
Source: EEGB stylised teaching data (data/). Not official statistics.

Fig S06-F06 · replicate: s06_figures.R / .py

WHAT TO SEE

- Each dot compares inflation now with inflation 12 months earlier.
- The fitted line: inflation now ≈ **0.74 + 1.13 ×** last year's rate.
- A slope near or above 1 means high inflation tends to **stay** high — or rise further.
- Why? Contracts, rents, wages and prices are set looking backward and forward.

Expectations Shift the Curve



Conceptual diagram.

Fig S06-F07 · replicate: s06_figures.R / .py

WHAT TO SEE

- The Phillips curve links inflation to **slack**: less spare capacity, more pressure on prices.
- But each curve is drawn for a given level of **expected** inflation.
- If people come to expect 30%, the curve shifts up: the same slack now comes with 30%-type inflation.
- So central banks fight hard to keep expectations **anchored** — even at the cost of growth.

Why People Expect What They Expect

Looking backward

Landlords, unions and suppliers ask: what did prices do last year? They build that into next year's rent, wages and price lists.

Looking at the central bank

If the CBN is believed when it says it will bring inflation down, people raise prices less. Credibility is a policy tool.

Looking at the naira

In Nigeria the exchange rate is a daily, visible signal. A falling naira quickly raises inflation expectations.

Common Mistakes When Reading Inflation

THE TRAP	THE BETTER HABIT
✗ “Inflation is falling, so prices are falling.”	✓ Falling inflation means prices rise **more slowly**. The price level is still higher.
✗ Treating every inflation episode as too much demand.	✓ Check **food vs core**, the exchange rate and activity before deciding.
✗ Comparing inflation across a **rebasing** as if nothing changed.	✓ Use the same base, or month-on-month changes on the new base.
✗ Assuming a rate rise brings inflation down within a quarter.	✓ Monetary policy works with **long lags** — often a year or more.
✗ Indexing a contract to 'inflation' without naming the index.	✓ Name the measure (headline, core) and the base year in the contract.

CONCEPT CHECK

Quick Check: What Would You Say?

?

Headline inflation is 30%, core is 22% and the trimmed mean is 23%. What is the best reading?

A. Underlying inflation is about 30%, so plan for 30% every year

B. A few items (likely food and energy) are pushing headline above the underlying trend of about 22–23%

C. Core inflation is wrong because it leaves out food

D. Inflation is under control because core is lower

Discuss with your neighbour for 60 seconds — answer and reasoning in the speaker notes.

Lab: Diagnose the 2023–24 Inflation Surge

YOU WILL NEED

Data
`data/ng_macro_monthly.csv`

R script
`sessions/s06/R/s06_lab.R`

Python script
`sessions/s06/python/s06_lab.py`

⏱ **20 minutes**

1. Compare headline, food, core and PMI **before and after** mid-2023.
2. Compute the **food minus core** gap by year.
3. Correlate inflation with naira depreciation at different **lags** — where is the link strongest?
4. Fit inflation now against inflation a year ago to measure **persistence**.
5. Check the **real policy rate**: were rates above or below inflation?

Deliverable: a three-sentence diagnosis — demand, supply or both — citing at least two numbers, and one implication for your pricing.

WORDS TO OWN

Today's Vocabulary in Plain English

Aggregate demand	Total spending on an economy's goods and services.
Aggregate supply	Total output firms will produce at each price level.
Demand shock	A change in spending that moves prices and output the same way.
Supply shock	A change in costs that moves prices and output in opposite ways.
Stagflation	High inflation combined with weak or falling output.
Core inflation	Inflation excluding volatile items such as farm produce and energy.
Trimmed mean	Average price change after dropping the most extreme movers.
Pass-through	How much of an exchange-rate change shows up in local prices.
Phillips curve	The link between economic slack and inflation.
Anchored expectations	People's belief that inflation will return to target.

KEY TAKEAWAYS

Session 6 in 5 Ideas

01 **AD–AS**: demand shocks move prices and output together; supply shocks move them apart.

02 Diagnose before you prescribe: rates cure **demand** inflation far better than **supply** inflation.

03 Use **headline** for the cost of living; **core and trimmed mean** for the underlying trend.

04 Watch for **rebasing** — a new ruler is not a new reality.

05 Inflation is **sticky** because of expectations; credibility is worth fighting for.



NEXT › Session 7: National Accounts

Before next session: Read the Mankiw chapters on measuring national income (GDP). Find your company's revenue for the last three years and ask: how much of the growth was real?